

Optimization Homework 6

Dai Xunlian 225040015
225040015@link.cuhk.edu.cn

Fall 2025

Q1

(a) Proof that f is not Lipschitz smooth

Problem: Show that $f(x) = |x|^{3/2}$ is not Lipschitz smooth.

Solution:

A function f is Lipschitz smooth if there exists a constant $L \geq 0$ such that $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all x, y . This condition implies that the magnitude of the second derivative (Hessian) is bounded by L everywhere.

Let us compute the derivatives of $f(x) = |x|^{3/2}$. The first derivative is:

$$f'(x) = \frac{3}{2} \operatorname{sgn}(x) |x|^{1/2}$$

The second derivative is:

$$f''(x) = \frac{3}{2} \cdot \frac{1}{2} |x|^{-1/2} = \frac{3}{4\sqrt{|x|}} \quad \text{for } x \neq 0$$

Now, consider the limit as x approaches 0:

$$\lim_{x \rightarrow 0} |f''(x)| = \lim_{x \rightarrow 0} \frac{3}{4\sqrt{|x|}} = +\infty$$

Since the second derivative is unbounded near $x = 0$, there exists no finite constant L such that $|f''(x)| \leq L$ for all x . Therefore, f is **not Lipschitz smooth**.

(b) Failure of Gradient Descent with Constant Step Size

Problem: Prove that for any constant step size α , there exists an x_0 such that $f(x_1) \geq f(x_0)$.

Solution:

The gradient descent update rule is $x_1 = x_0 - \alpha f'(x_0)$. Assume we choose an initial point $x_0 > 0$. Then $f'(x_0) = \frac{3}{2} x_0^{1/2}$.

$$x_1 = x_0 - \alpha \frac{3}{2} x_0^{1/2}$$

For the algorithm to fail to decrease the objective function, we need $f(x_1) \geq f(x_0)$. Since $f(x) = |x|^{3/2}$ is symmetric and monotonically increasing with $|x|$, this implies $|x_1| \geq |x_0| = x_0$. Since we are subtracting a positive term from x_0 , $x_1 < x_0$. Thus, for $|x_1| \geq x_0$, we must have $x_1 \leq -x_0$.

Substituting the expression for x_1 :

$$x_0 - \frac{3}{2} \alpha x_0^{1/2} \leq -x_0$$

$$2x_0 \leq \frac{3}{2} \alpha x_0^{1/2}$$

Dividing by $x_0^{1/2}$ (since $x_0 > 0$):

$$2x_0^{1/2} \leq \frac{3}{2} \alpha \implies x_0^{1/2} \leq \frac{3}{4} \alpha \implies x_0 \leq \frac{9}{16} \alpha^2$$

Conclusion: For any fixed $\alpha > 0$, if we choose an initial point $x_0 \in (0, \frac{9}{16} \alpha^2]$, the gradient step will overshoot the origin significantly such that $|x_1| \geq x_0$, and thus $f(x_1) \geq f(x_0)$.

Q2

Problem: Minimize $f(x) = x^\top Ax$ with $A = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$ starting from $x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Note that $\nabla f(x) = 2Ax = \begin{bmatrix} 4x_1 \\ 10x_2 \end{bmatrix}$. At x_0 , $\nabla f(x_0) = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$.

(a) Manual Calculations (One Step)

1. Constant Step Size ($\alpha = 0.1$):

$$x_1 = x_0 - 0.1\nabla f(x_0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.4 \\ 1.0 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0 \end{bmatrix}$$

2. Exact Line Search: We minimize $\phi(\alpha) = f(x_0 - \alpha\nabla f(x_0))$. For a quadratic form, the optimal step size is:

$$\alpha^* = \frac{\nabla f(x_0)^\top \nabla f(x_0)}{2\nabla f(x_0)^\top A \nabla f(x_0)}$$

Numerator: $4^2 + 10^2 = 16 + 100 = 116$.

Denominator term: $g^\top Ag$ where $g = [4, 10]^\top$.

$$g^\top Ag = 2(4)^2 + 5(10)^2 = 32 + 500 = 532$$

$$\alpha^* = \frac{116}{2(532)} = \frac{116}{1064} \approx 0.1090$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.1090 \begin{bmatrix} 4 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 - 0.436 \\ 1 - 1.090 \end{bmatrix} = \begin{bmatrix} 0.564 \\ -0.090 \end{bmatrix}$$

3. Backtracking Line Search: Parameters: $\gamma = 0.5, \sigma = 0.2$. Condition: $f(x_{new}) \leq f(x) - \gamma\alpha\|\nabla f(x)\|^2$. Limit: $7 - 0.5 \cdot \alpha \cdot 116 = 7 - 58\alpha$.

- **Try $\alpha = 1$:** $x_{new} = [-3, -9]$. $f(new) = 2(9) + 5(81) = 423$. Limit = -51 . Fail.
- **Try $\alpha = 0.2$:** $x_{new} = [0.2, -1]$. $f(new) = 2(0.04) + 5(1) = 5.08$. Limit = $7 - 11.6 = -4.6$. Fail.
- **Try $\alpha = 0.04$:** $x_{new} = [0.84, 0.6]$. $f(new) = 2(0.84^2) + 5(0.6^2) = 1.41 + 1.8 = 3.21$. Limit = $7 - 2.32 = 4.68$. **Pass.**

Result: $x_1 = \begin{bmatrix} 0.84 \\ 0.60 \end{bmatrix}$.

(b) Python Implementation Results

The gradient descent algorithm was implemented in Python for 10 iterations using the three strategies. The objective function values $f(x_k)$ are recorded below.

Table 1: Objective Function Values $f(x_k)$ over 10 Iterations

Iteration	Constant ($\alpha = 0.1$)	Exact Line Search	Backtracking
0	7.0000	7.0000	7.0000
1	0.7200	0.6767	3.2112
2	0.2592	0.0654	1.4729
3	0.0933	0.0063	0.6756
4	0.0336	0.0006	0.3099
5	0.0121	0.0001	0.1421
6	0.0044	0.0000	0.0652
7	0.0016	0.0000	0.0299
8	0.0006	0.0000	0.0137
9	0.0002	0.0000	0.0063
10	0.0001	0.0000	0.0039

Q3

Problem: $f(x, y) = \frac{3}{4}x^{4/3} + \frac{1}{2}(x + y)^2$.

(a) Global Minimizer

The function is the sum of two non-negative terms: $\frac{3}{4}(x^2)^{2/3} \geq 0$ and $\frac{1}{2}(x+y)^2 \geq 0$. The global minimum occurs when both terms are zero: 1. $x = 0$ 2. $x + y = 0 \implies 0 + y = 0 \implies y = 0$.

Global Minimizer: $(x^*, y^*) = (0, 0)$.

(b) Newton's Method Divergence

Gradient: $\nabla f = \begin{bmatrix} x^{1/3} + (x+y) \\ x+y \end{bmatrix}$. Hessian: $H = \begin{bmatrix} \frac{1}{3}x^{-2/3} + 1 & 1 \\ 1 & 1 \end{bmatrix}$. Newton step: $\Delta = -H^{-1}\nabla f$. The inverse Hessian is $H^{-1} = 3x^{2/3} \begin{bmatrix} 1 & -1 \\ -1 & \frac{1}{3}x^{-2/3} + 1 \end{bmatrix}$. Calculating the step for x :

$$\Delta_x = -[(H^{-1})_{11}(\partial_x f) + (H^{-1})_{12}(\partial_y f)]$$

$$\Delta_x = -[3x^{2/3}(x^{1/3} + x + y) - 3x^{2/3}(x + y)] = -[3x + 3x^{2/3}(x + y) - 3x^{2/3}(x + y)] = -3x$$

Update rule:

$$x_{k+1} = x_k + \Delta_x = x_k - 3x_k = -2x_k$$

Starting from any $x_0 \neq 0$, the sequence is $x_k = (-2)^k x_0$. Since $|-2| > 1$, $|x_k| \rightarrow \infty$ as $k \rightarrow \infty$. Thus, Newton's method diverges.

(c) Projected Gradient Descent

Constraint $\Omega = \{(x, y) : x + y = 1\}$. Let the current point be $z_k = (x_k, y_k)$. Step 1 (Unconstrained Gradient Step):

$$\hat{z} = z_k - \alpha_k \nabla f(z_k) = \begin{bmatrix} x_k - \alpha_k(x_k^{1/3} + x_k + y_k) \\ y_k - \alpha_k(x_k + y_k) \end{bmatrix}$$

Step 2 (Projection): The projection of a point $\hat{z}$ onto the line $a^\top z = b$ (where $a = [1, 1]^\top, b = 1$) is:

$$z_{k+1} = \hat{z} - \frac{a^\top \hat{z} - b}{\|a\|^2} a$$

$$x_{k+1} = \hat{x} - \frac{\hat{x} + \hat{y} - 1}{2}, \quad y_{k+1} = \hat{y} - \frac{\hat{x} + \hat{y} - 1}{2}$$

Q4

(a) Projection onto Line L

Let $L = \{p + tv \mid t \in \mathbb{R}\}$. We minimize $D(t) = \|(p + tv) - x\|^2$.

$$D(t) = \|tv - (x - p)\|^2 = t^2\|v\|^2 - 2tv^\top(x - p) + \|x - p\|^2$$

Setting $D'(t) = 0$:

$$2t\|v\|^2 - 2v^\top(x - p) = 0 \implies t^* = \frac{v^\top(x - p)}{\|v\|^2}$$

Thus, $\mathcal{P}_L(x) = p + \frac{v^\top(x-p)}{\|v\|^2}v$.

(b) Projection onto Intersection

We want to project x onto $C = L \cap B$. If $\mathcal{P}_L(x) \in B$, then $\mathcal{P}_L(x) \in L \cap B = C$. By definition, $\mathcal{P}_L(x)$ is the unique point in L closest to x . Since $C \subset L$, $\mathcal{P}_L(x)$ is also the point in C closest to x . Therefore, if the projection onto the line is feasible (lies within the ball), it is also the projection onto the intersection.

$$\mathcal{P}_C(x) = \mathcal{P}_L(x) \quad \text{if } \mathcal{P}_L(x) \in B$$